

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT, open to SF/NYC/Seattle relocation

PROFESSIONAL SUMMARY

New-grad applied AI engineer (Master's CS, Kent State, May 2025) with shipped production full-stack code, not just course projects. Shipped **Manga Lens to the Chrome Web Store** integrating four AI vision providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local) behind a single TypeScript interface with provider-specific payload handling for 29 sites - the same multi-API integration discipline customer-embedded FDE work depends on. Shipped a FastAPI + React/TypeScript creative platform (Dream Decoder). Built sensor + hardware data pipelines (IoT Arduino + DHT11 + DS3231 with walk-forward time-series forecasting). Production AI work at Progress Solutions: healthcare RAG (~35% retrieval precision), agentic workflows, FastAPI/Docker services. Open to relocating to San Francisco, New York, or Seattle. US-based, F-1 OPT.

CORE COMPETENCIES

Full-Stack Web Development • Multi-API Integration • Customer-Iterative Delivery • Data Pipelines • Sensor & Hardware Data (IoT) • Production Shipping at Scale • Python • React • TypeScript • FastAPI • Docker • PostgreSQL

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present | USA

Jr. AI/ML Engineer Trainee - Production AI, Full-Stack Integration

- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects ~**30%**.
- Built **RAG systems** for clinical knowledge retrieval; semantic chunking + transformer embeddings improved contextual retrieval precision ~**35%**, reduced irrelevant retrieval >**30%**.
- Built **agentic LLM workflows** with structured reasoning and tool discipline; improved agent response stability ~**25%**.
- Built preprocessing pipelines (Pandas, NumPy) processing **EHR extracts, appointment histories, support ticket logs**; raised dataset reliability above **98%** and cut downstream model instability ~**40%**.
- Shipped predictive ML (scikit-learn, XGBoost); recall **15-20%** gain on high-risk categories while holding precision >**90%**.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023 | India (10 months CS internship)

Database & DevOps Performance Engineer (Intern) - Enterprise Data Pipelines

- Built **CI/CD pipelines (Jenkins)** for SQL and stored-procedure deploys with release validation + rollback checkpoints; reduced deploy errors >**30%**, improved release cycle ~**35-40%** - the same shipping discipline customer-facing FDE work depends on.
- Tuned **T-SQL stored procedures** on Synthesis Order-to-Cash ERP modules; cut query execution time ~**20%** and data retrieval latency ~**25%**.
- Designed **SQL Server DMV + Grafana** performance dashboards; cut incident recurrence ~**25%**, improved RCA speed ~**30%**.
- Supported **RBAC + audit logging** for financial modules in compliance-sensitive oil & gas environment.

Suvidha Foundation

Jan 2022 - Mar 2022 | India

Machine Learning Engineer - Applied NLP, Document Q&A RAG

- Built a transformer-based summarization system over **5,000+ recorded sessions**; hierarchical long-context summarization cut manual review time **60-70%**.
- Implemented **document Q&A with RAG** + Flask API with lightweight web UI for non-technical staff - customer iteration with non-tech users.

PROJECTS

Manga Lens - Chrome Extension Integrating 4 AI Vision Providers (shipped, Chrome Web Store)

Full-stack browser extension built independently. Manifest V3, content scripts, service workers, captureVisibleTab for panel capture. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local) behind one TypeScript interface with **provider-specific payload handling**. Multi-section capture pipeline handles tall webtoon panels via viewport-slice screenshots with coordinate remapping. Seven-day translation cache, per-domain selector configs for **29 manga/webtoon sites**, privacy policy, narrowed host permissions.

TypeScript, React-adjacent, multi-API integration, shipping at scale

Dream Decoder - Full-Stack Multimodal Platform (FastAPI + React/TS)

Full-stack **FastAPI + React/TypeScript/Vite/Tailwind** app coordinating multimodal APIs (Perplexity Sonar + Replicate). Introduced **intermediate structured prompt transformation layers** - improved contextual alignment **~30%** and first-pass image success **~25-30%** over naive direct-prompt orchestration.

FastAPI, React, TypeScript, Vite, Tailwind, multi-API orchestration

IoT + ML Smart Building Indoor Temperature Prediction (Hardware + Sensor Data Pipeline)

Arduino + DHT11 + DS3231 sensor pipeline with **PLX-DAQ logging**; time-series forecasting with **walk-forward validation** comparing Random Forest, SVR, and Linear Regression. Lag-window feature engineering ($n_{in}=6$); Random Forest achieved **~20-30% lower MAE** vs linear baseline. Hardware-to-data-processing pipeline end-to-end.

Arduino C++, DHT11, DS3231, sensor data pipelines, walk-forward time series

Geospatial Road Network Analysis (Large-Volume Data Processing)

Parsed raw **OpenStreetMap XML** into a weighted NetworkX DiGraph; Haversine-based edge weighting; Dijkstra shortest-path routing; **betweenness centrality on a 20k-node subgraph**; Louvain community detection; Minimum Spanning Tree backbone analysis. **~30%** runtime reduction via subgraph scoping.

NetworkX, Python, geospatial, large-scale graph processing

Driver Drowsiness Detection - Real-Time YOLOv8

Unified **YOLOv8** detection-and-classification model replacing two-stage CNN; reduced inference latency **~30%**. Sliding-window confidence aggregation suppressed blink-driven false positives (**~25% reduction**).

YOLOv8, OpenCV, real-time CV

EDUCATION

Master's Degree, Computer Science - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: Python, TypeScript, React, FastAPI, Flask, Node.js, C++ (Arduino), SQL (PostgreSQL, T-SQL, SQLite)

Full-Stack & Integration: Multi-API integration, RESTful APIs, full-stack web apps, customer-iterative delivery, Vite, Tailwind

Data & Pipelines: Pandas, NumPy, time-series forecasting, walk-forward validation, NetworkX, large-volume data processing

Infra & DevOps: Docker, Jenkins, CI/CD, Grafana, SQL Server DMVs, observability/logging

AI / ML: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, YOLOv8, RAG, agentic workflows, LangChain